Abtin Aryanpour

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Education

University of Waterloo Waterloo, ON

Candidate for BASc in Computer Engineering, Co-op Program *Expected Apr. 2031*

- Coursework: Fundamentals of Programming (C++), Linear Algebra, Calculus, Mechanics, Engineering Design

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: Django, REST APIs, PyTorch, OpenCV, NumPy, pandas, scikit-learn, Streamlit
Tools & Platforms: Git, GitHub Actions (CI), pytest, CMake, SQLite, AWS CloudFront

Projects

Andon: Machine-Sound Anomaly Detection | *Python, PyTorch, C, Pico 2* | GitHub Oct. 2026 – Present

- Built a machine-sound anomaly detector trained only on healthy audio: a **98k-weight CNN** reached **92.13% mean AUC** on the held-out DCASE 2020 evaluation set vs. 79.70% for the reference autoencoder (3 seeds)
- Reproduced the published DCASE baseline in PyTorch within **0.88 AUC points** on all 6 machine types, and released checkpoints so one command re-checks the held-out CNN results from a fresh clone
- Turned scores into a live andon light with SPC control limits, hysteresis and a Streamlit dashboard; on held-out machines it caught **65.57% of 30 s faults**, on average **10.57 s** after onset, at 5.53 false alarms per hour
- Quantized the CNN to int8 for a Cortex-M33 with a custom fixed-point reference: **98.7 KB** (from 383.0 KB) at **92.08% held-out AUC**, exporting C headers and test vectors in CI; now porting to a Raspberry Pi Pico 2

Image Stitcher, Waterloo Aerial Robotics Group | *Python, PyTorch, OpenCV* | PR #215 Oct. 2026 – Present

- Developed a groundside orthomosaic stitcher on the autonomy subteam, merging out-of-order, GPS-tagged photos from **3 drones** into one georeferenced map using XFeat learned features and RANSAC
- Fused images with noisy GPS to recover heading, aligning photos within **6.9 cm** at 3 m simulated GPS error
- Designed a bundle-adjustment-style least-squares solver, cutting simulated misalignment from **13–41 to 4–7 cm**
- Handled 180-degree turns and repetitive terrain, forming one map on **10/10** simulated 91-photo missions
- Added lens undistortion and blur, exposure and tilt checks; built a ground-truth mission simulator and **80 tests**

Doppel.exe: Continuous Authentication | *Python, AES-GCM, Windows APIs* | GitHub Sep. – Oct. 2026

- Built a privacy-preserving Windows agent that scores typing rhythm, storing only encrypted timings, never letters
- Cut mean EER **7x** (0.164 to 0.023) on 500 held-out Aalto dataset users by log-scaling timings and capping outliers
- Reproduced the CMU keystroke benchmark within **0.0002 mean EER** across 3 detectors before building on it

CyberShield: Full-Stack Discussion Forum | *Python, Django, JavaScript, Tailwind CSS* | GitHub Mar. 2025

- Sole developer of a responsive forum where users post, comment and vote, with sorting, topic filtering and fuzzy search (Fuse.js), built on Django, Django REST Framework and SQLite
- Secured accounts with Django authentication, CSRF protection and reCAPTCHA on login and signup; added rich-text posts (CKEditor) and profile picture uploads
- Deployed live behind AWS CloudFront with a seeded demo dataset; won **1st place** in Coding & Programming at the Canadian National Leadership Conference 2025 (32 teams)

Experience

Homework Helper & Assistant Teacher Sep. 2022 – Jul. 2025

Spirit of Math Schools Inc. *Richmond Hill, ON*

- **Homework Helper (2024–2025):** Ran a drop-in homework help classroom as one of two staff, triaging questions from students ranging from kindergarten to Grade 11 and diagnosing where each one was stuck
- **Assistant Teacher (2022–2024):** Supported classes of 16 students, teaching lessons and contest-style algebra

Awards & Achievements

1st Place (2025, CNLC) and 4th Place (2024, FBLA): Coding & Programming, national competitions
Euclid (86) and Hypatia (33) Honour Roll; Fermat and CSMC Distinction: University of Waterloo, 2025–2026
Excellence in Mathematics: Bayview Secondary School Mathematics Department (2025, 2026)